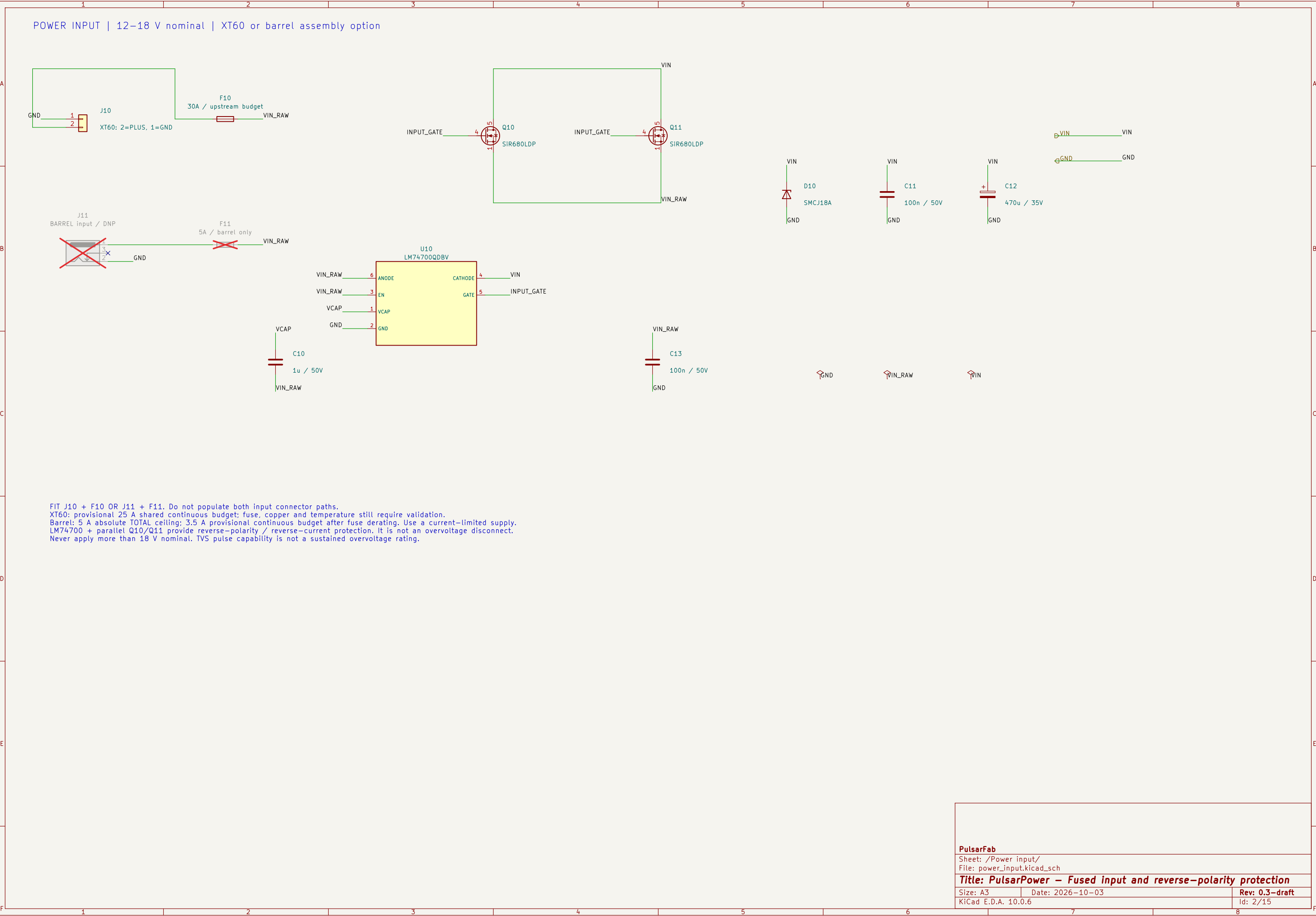


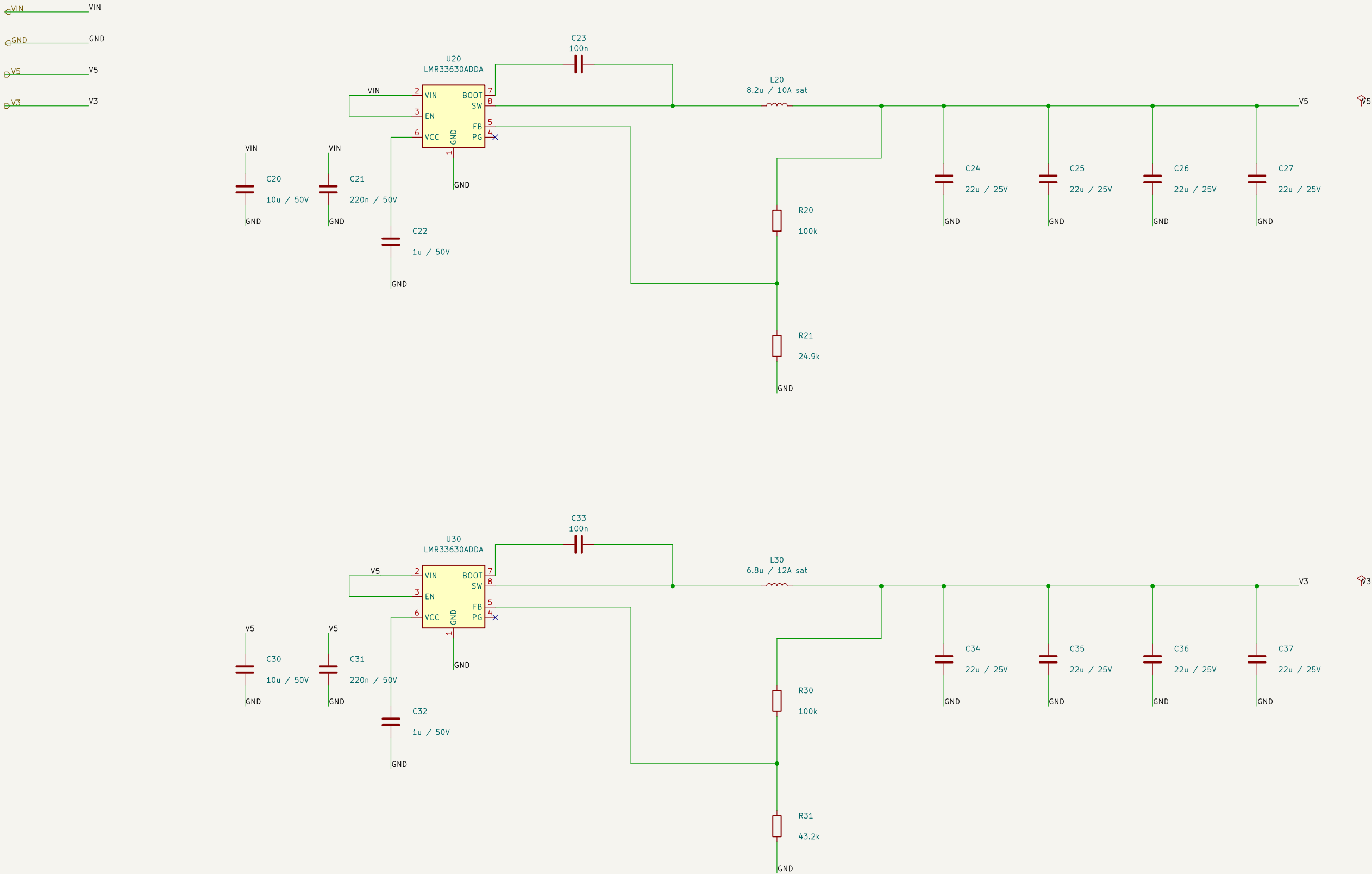
PULSARPOWER | 2 heaters + 4 monitored DC outlets + 4 USB 2.0 ports

PROTOTYPE FLOORPLAN – routing, full-load thermal validation and new firmware are still required.





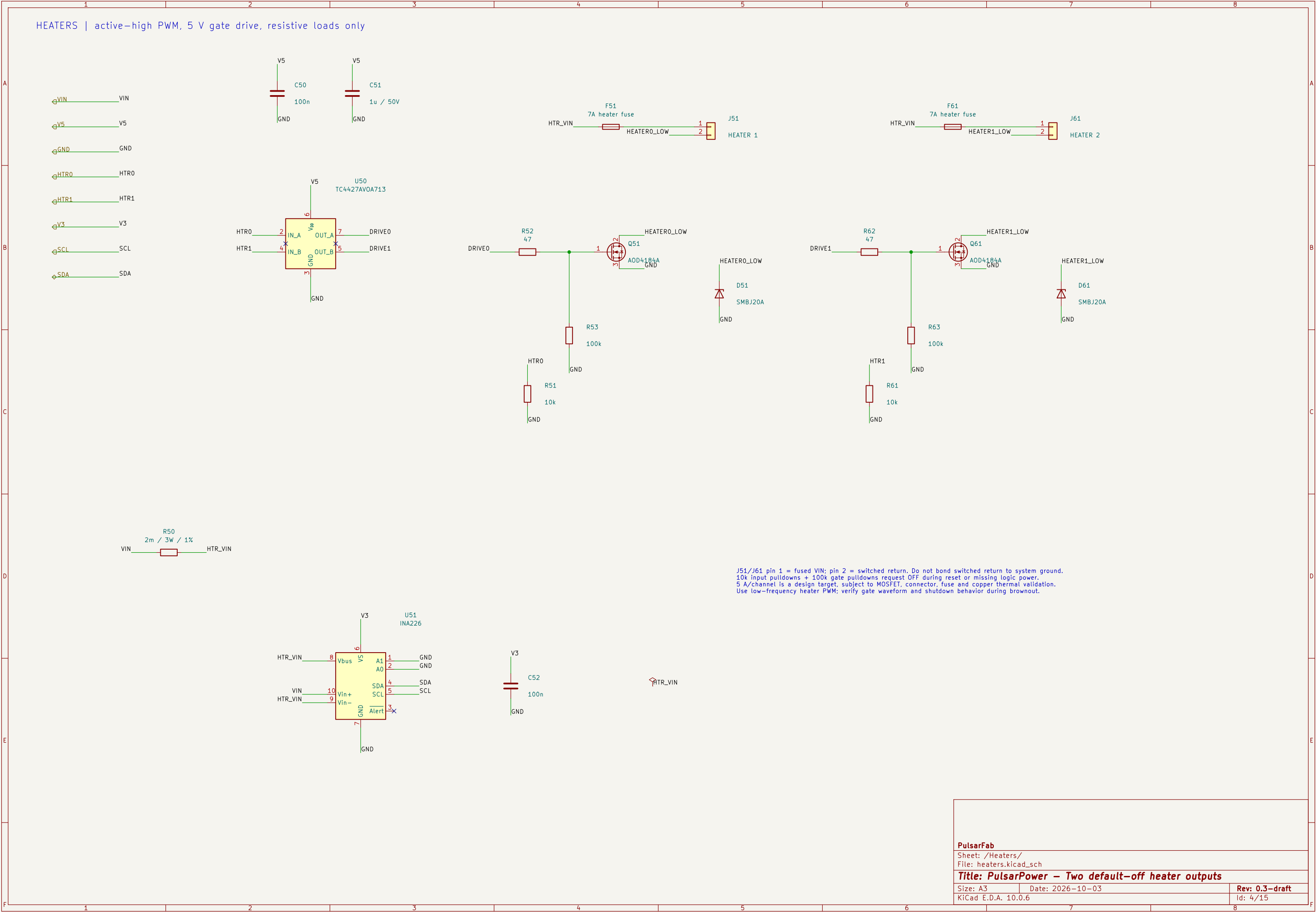
DC SUPPLIES | LMR33630ADDA, 400 kHz | 5 V / 3 A + 3.3 V logic

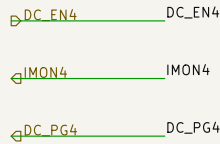
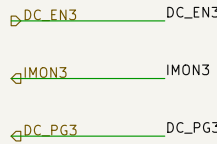
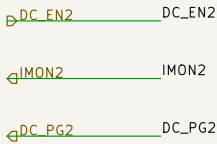
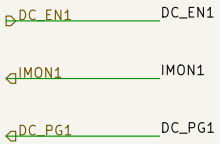
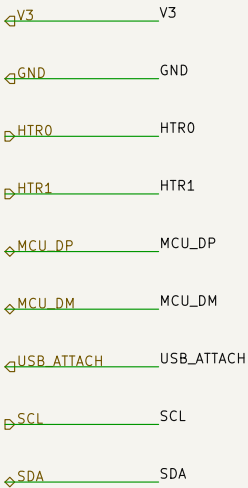
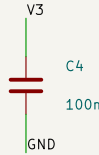
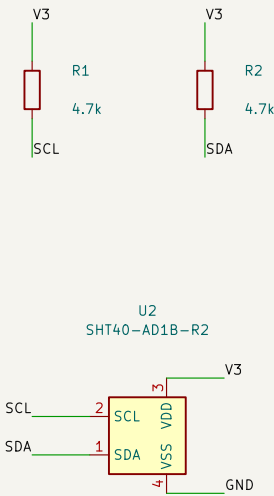
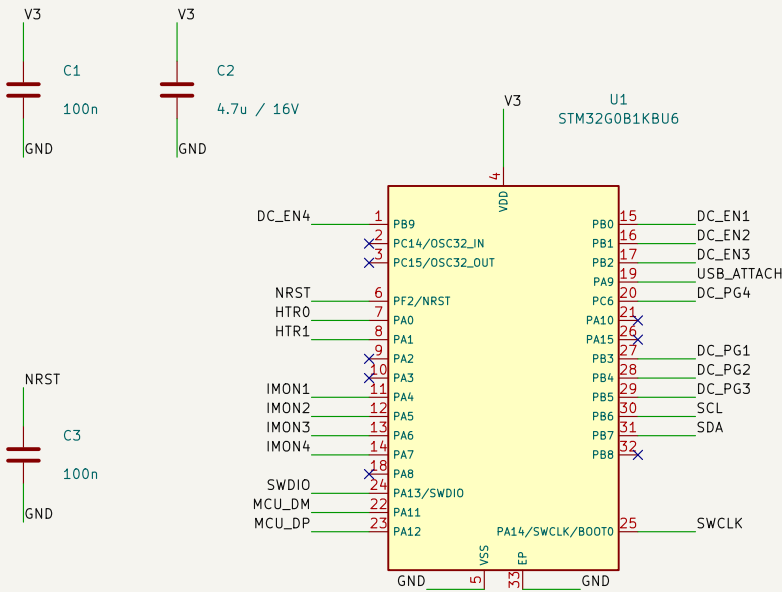


Follow TI SNVSN3F Table 9-2. 5 V uses nearest preferred 8.2 uH to the 8 uH example.
Place ceramic input/boot/VCC capacitors at the IC; keep SW copper compact and FB away from SW.
USB budget: 4 x 500 mA plus about 0.4 A for logic at full activity. Verify inductor saturation, MLCC bias, transient response and regulator temperature before routing release.

PulsarFab		
Sheet: /DC supplies/ File: supplies.kicad_sch		
Title: PulsarPower – 5 V USB and 3.3 V logic supplies		
Size: A3	Date: 2026-10-03	Rev: 0.3-draft
KiCad E.D.A. 10.0.6		Id: 3/15

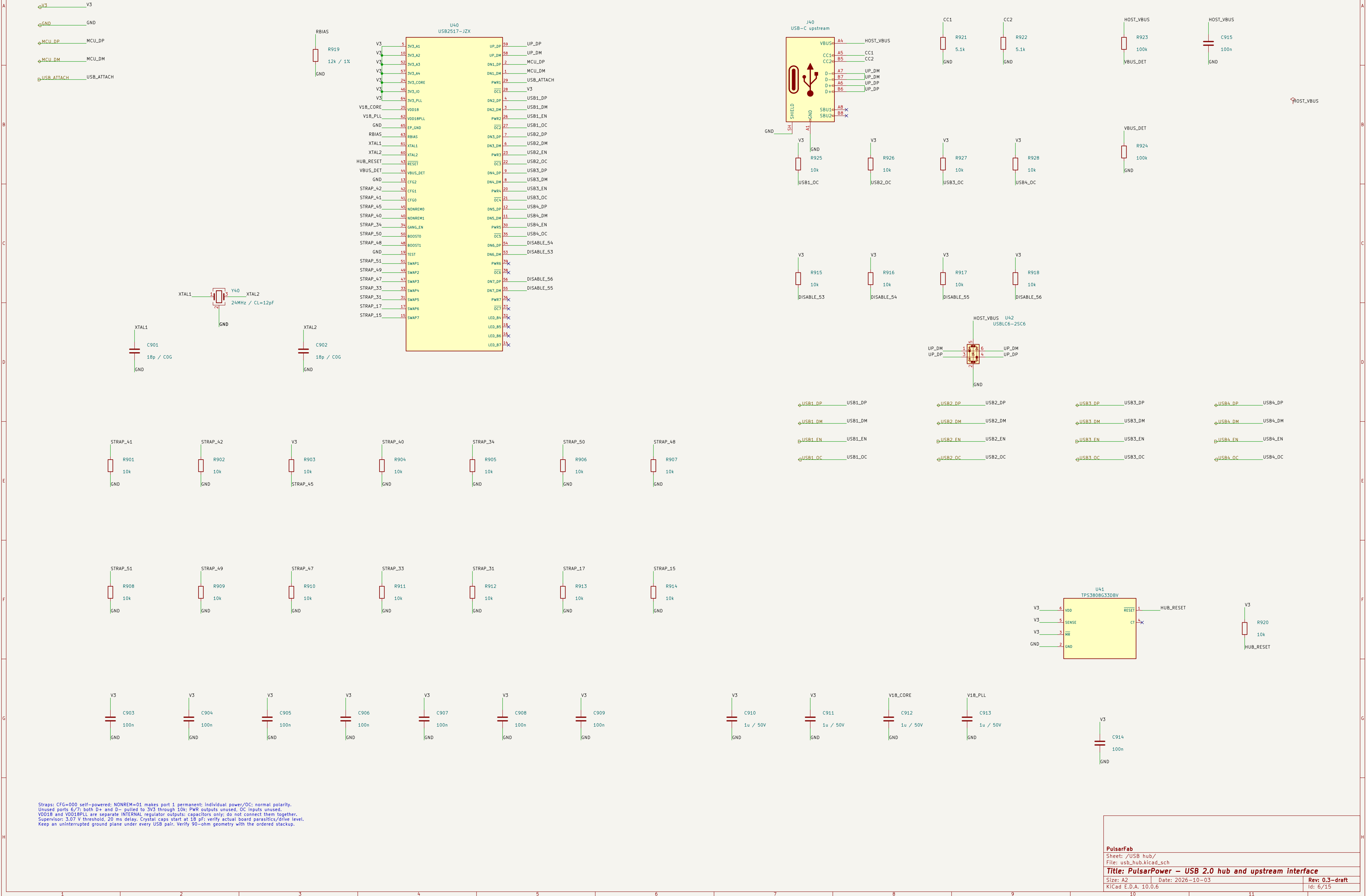
HEATERS | active-high PWM, 5 V gate drive, resistive loads only





PA0/PA1: active-high TIM2 heater drive. PB0/PB1/PB2/PB9: static DC GPIO only (never PWM).
PA4..PA7: ADC IMON1..4. PB3/PB4/PB5/PC6: active-low DC faults.
PA9: internal hub port 1 enable (3.3 V). Attach USB only while high; disable UCPD dead-battery pull-down.
SHT40 at 0x44; keep physically away from power components. New firmware pin map is required.

USB HUB | USB2517 | internal port 1 + external ports 2–5; ports 6/7 disabled



SWITCHED DC OUTLET | 12–18 V / 5 A service | hardware voltage inhibit / latched breaker

Legend:

- VIN: VIN
- V3: V3
- GND: GND
- ENABLE: ENABLE
- CURRENT: CURRENT
- PGOOD: PGOOD

U102 TVS1800DRV

- VIN (4) IN
- VIN (5) IN
- VIN (6) IN
- GND (1) GND
- GND (2) GND
- GND (3) GND
- EP_GND (7) GND

U101 TPS25974LRPW

- VIN (5) IN
- EN_SW (1) EN_UVLO
- OV_SENSE (2) OVLO
- GND (8) GND
- OUT (6) OUT
- PG (3) PGOOD
- PGTH (4) PG_SENSE
- ILM_MON (9) IMON_RAW
- dVdT (7) DVDT
- ITIMER (10) ITIMER

Components:

- C101: 100n / 50V
- C102: 100n / 50V
- C103: 4.7n / 50V
- C104: 10n
- R101: 100k
- R102: 20k
- R103: 150k
- R104: 10k
- R105: 10k
- R106: 10k
- R107: 1k / 1%
- R109: 10k
- R110: 1k
- J101: 5A DC / center +
- D101: S554

Notes:

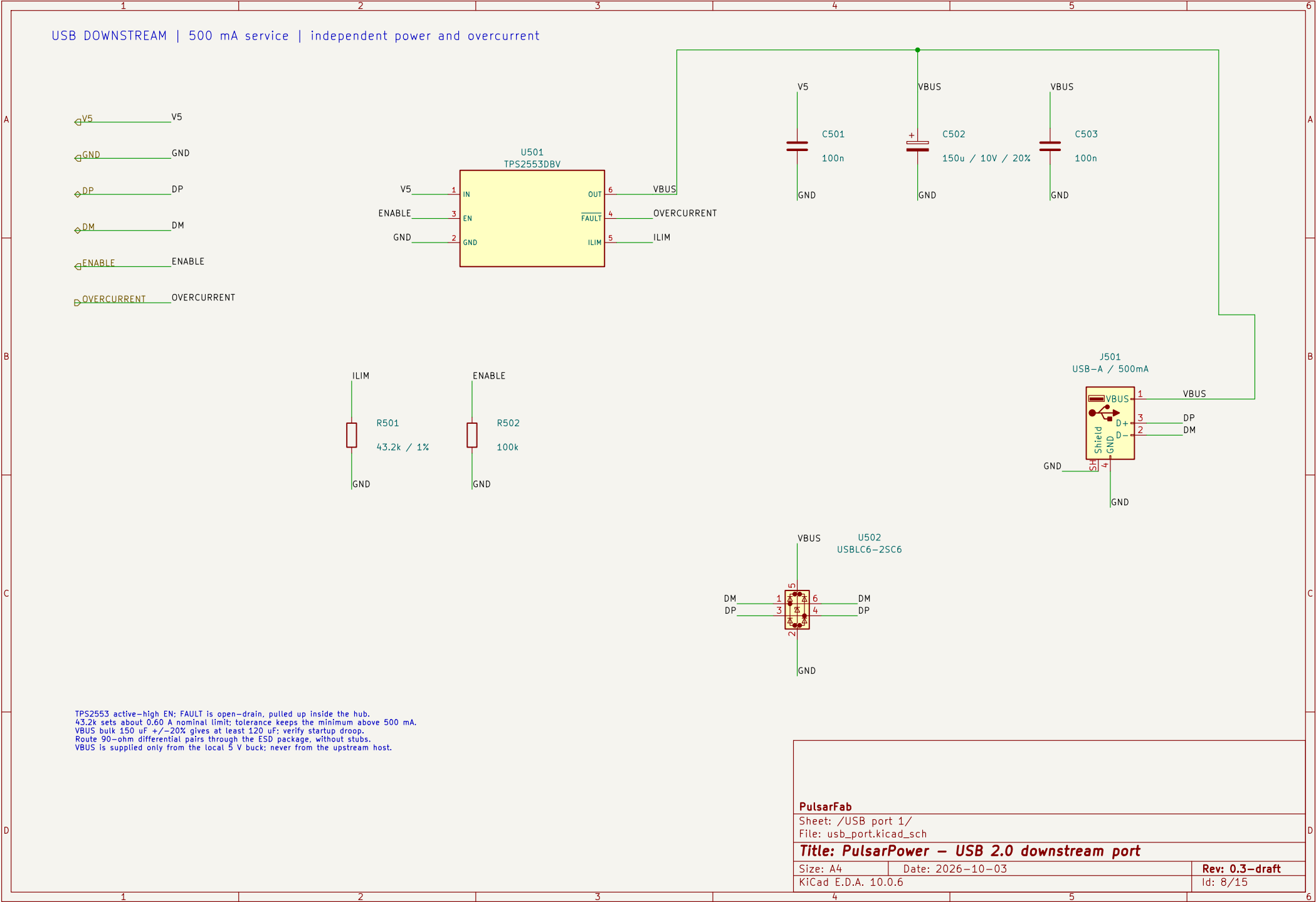
- TPS25974L: 1k sets 5.75 A nominal breaker threshold for 5 A service. ITIMER open; latch–off.
- ILM is BOTH current setting and monitoring: $105.5 \mu\text{A/A} \times 1\text{k} = 0.1055 \text{ V/A}$; 0.528 V at 5 A.
- 10k/10n filter isolates ADC capacitance from ILM. Calibrate channels; do not attach a capacitor directly to ILM.
- Independent local OVLO: $150\text{k}/10\text{k} \rightarrow 19.2 \text{ V}$ nominal. PG threshold: $100\text{k}/20\text{k} \rightarrow 7.2 \text{ V}$ nominal.
- Positive line switched; ground permanently connected. No reverse blocking; do not backfeed outputs.
- IN pad5 / OUT pad6 are power/thermal rails, NOT GND. Reserve copper and thermal–via space.
- TVS1800 flat clamp replaces SMBJ18A: 24.7 V max at specified 35 A / 8–20 us / 125 C pulse; verify loop overshoot.

Project Information:

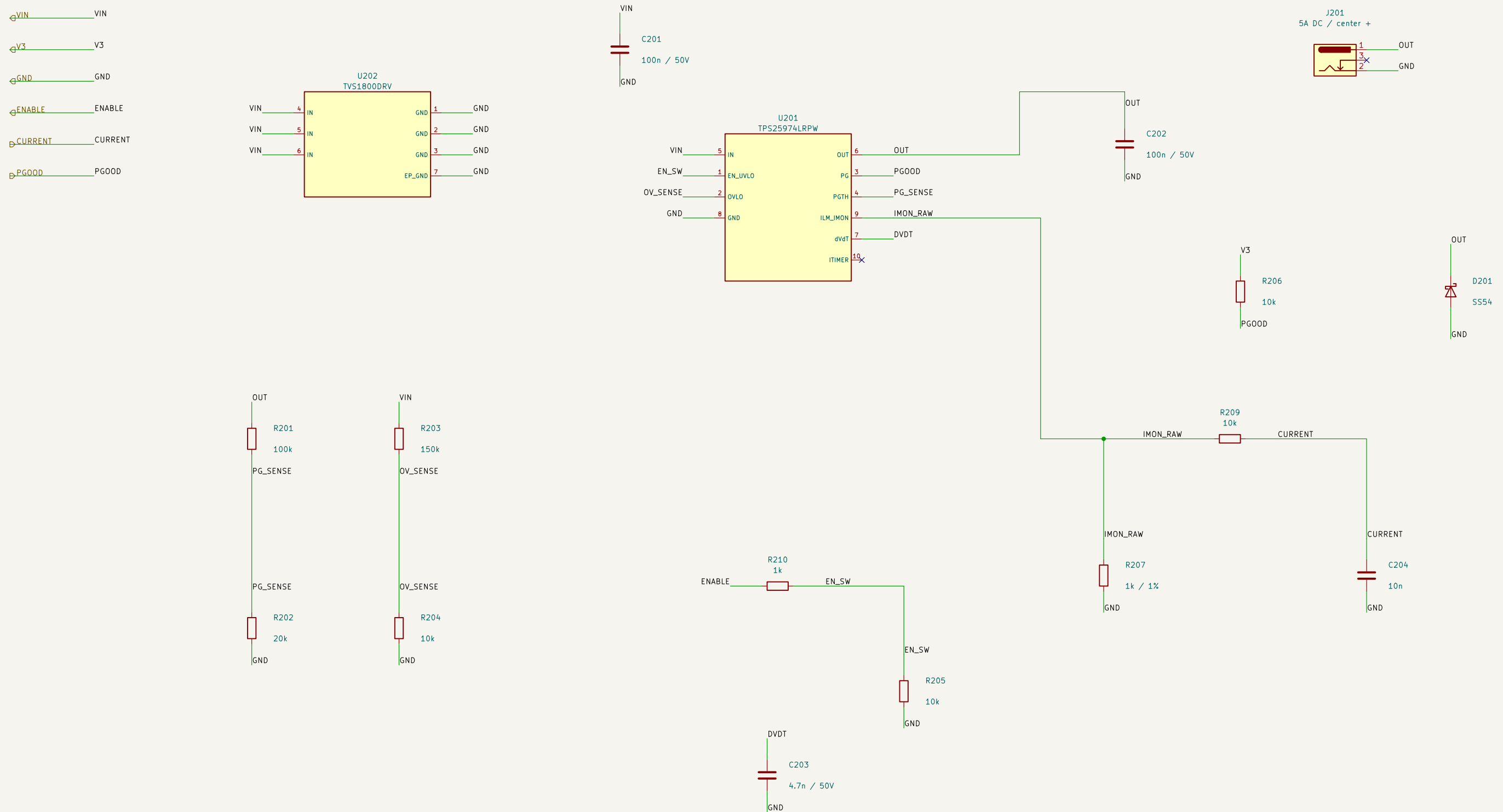
- Sheet: /DC outlet 1/
- File: dc_outlet.kicad_sch
- Title: PulsarPower – 5 A switched DC outlet
- Size: A3
- Date: 2026–10–03
- Rev: 0.3–draft
- KiCad E.D.A. 10.0.6
- Id: 7/15

TPS25974L: 1k sets 5.75 A nominal breaker threshold for 5 A service. ITIMER open: latch-off. ILM is BOTH current setting and monitoring: 105.5 uA/A x 1k = 0.1055 V/A. 0.528 V at 5 A. 10k/10n filter isolates ADC capacitance from ILM. Calibrate channels: do not attach a capacitor directly to ILM. Independent local OVLO: 150k/10k -> 19.2 V nominal. PG threshold: 100k/20k -> 7.2 V nominal. Positive line switched: ground permanently connected. No reverse blocking: do not backfeed outputs. In pads / OUT pads are power/thermal rails. NOT GND. Reserve copper and thermal-via space. TVS1800 flat clamp replaces SMBJ18A: 24.7 V max at specified 35 A - 8/20 us / 125 C pulse; verify loop overshoot.

USB DOWNSTREAM | 500 mA service | independent power and overcurrent

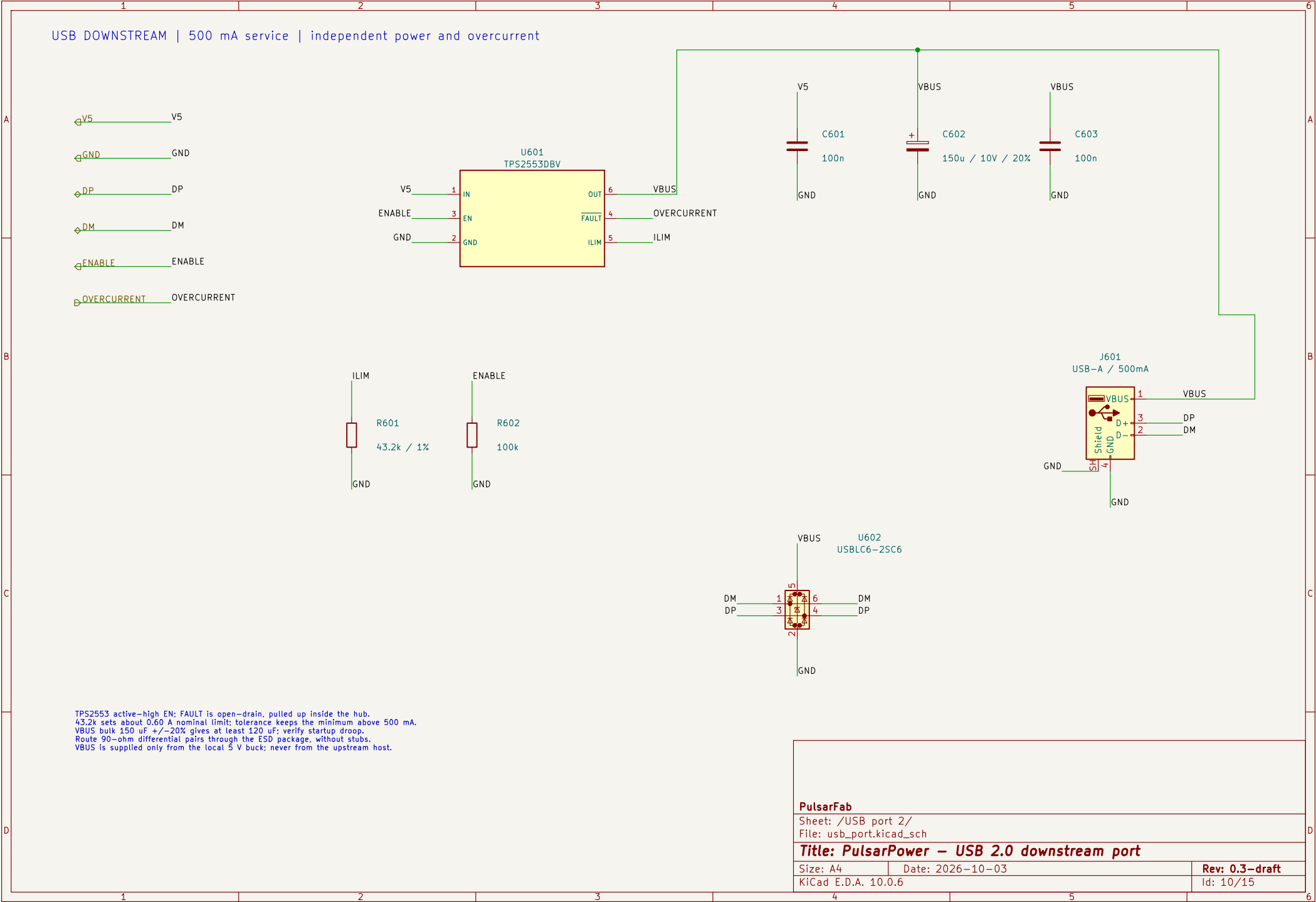


SWITCHED DC OUTLET | 12-18 V / 5 A service | hardware voltage inhibit / latched breaker

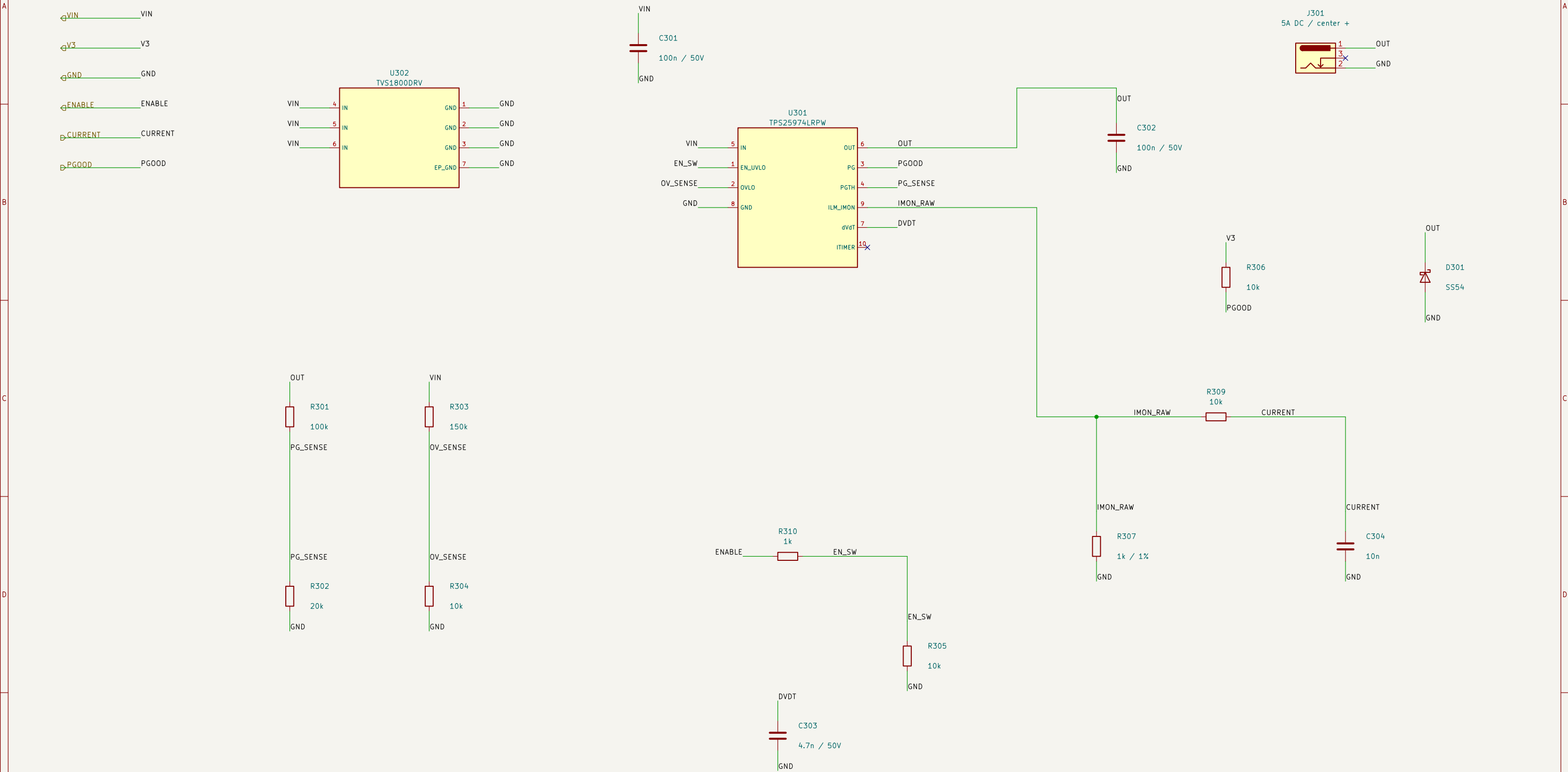


TPS25974L: 1k sets 5.75 A nominal breaker threshold for 5 A service. ITIMER open; latch-off. ILM is BOTH current setting and monitoring: $105.5 \text{ uA/A} \times 1k = 0.1055 \text{ V/A}$. 0.528 V at 5 A. 100k/10n filter isolates ADC capacitance from ILM. Calibrate channels; do not attach a capacitor directly to ILM. Independent local OVLO: 150k/10k $\rightarrow$ 19.2 V nominal. PG threshold: 100k/20k $\rightarrow$ 7.2 V nominal. Positive line switched; ground permanently connected. No reverse blocking; do not backfeed outputs. In pad5 / OUT pad6 are power/thermal; NOT GND. Reserve copper and thermal-via space. TVS1800 flat clamp replaces SMBJ18A: 24.7 V max at specified 35 A / 8–20 us / 125 C pulse; verify loop overshoot.

USB DOWNSTREAM | 500 mA service | independent power and overcurrent

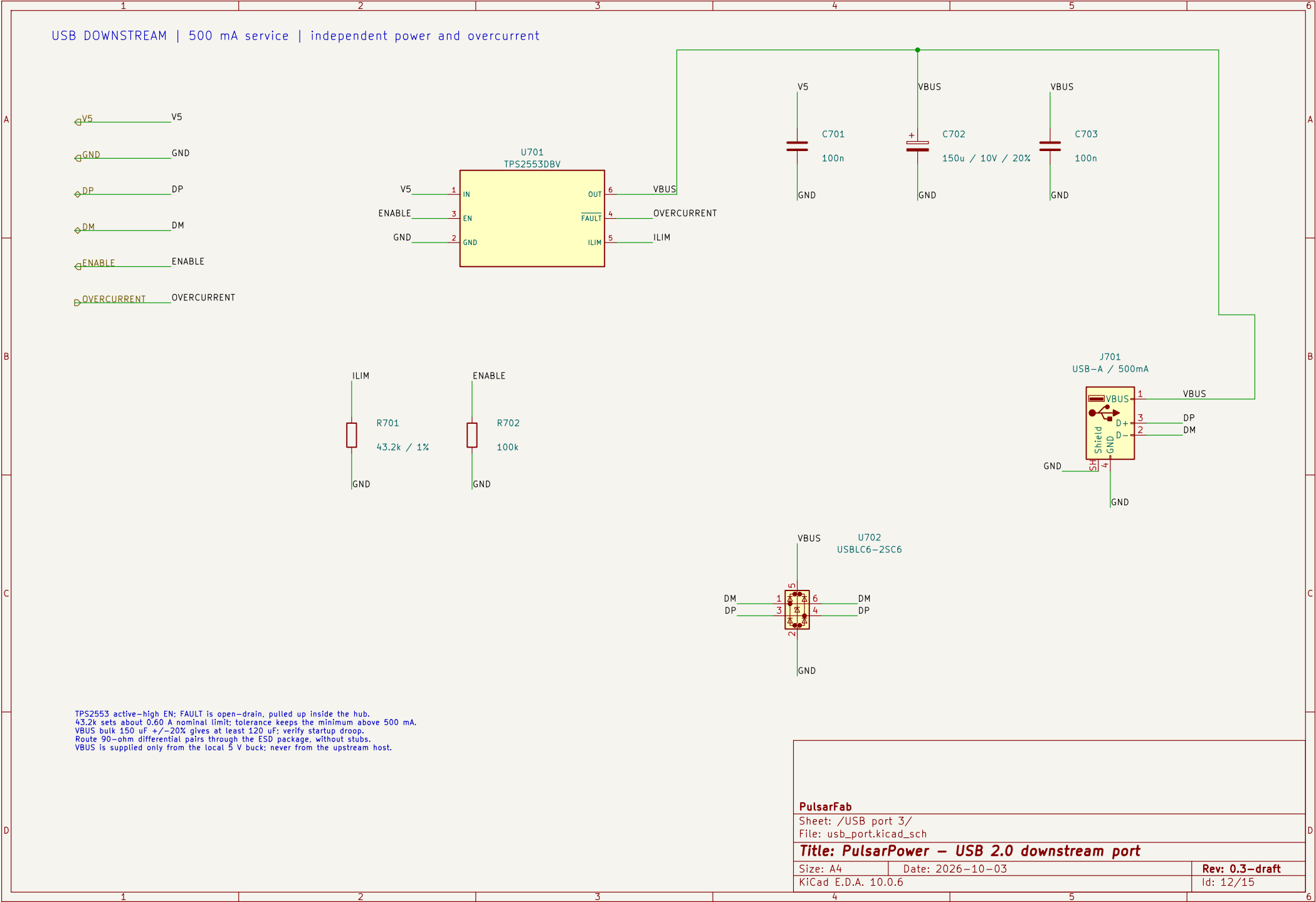


SWITCHED DC OUTLET | 12–18 V / 5 A service | hardware voltage inhibit / latched breaker

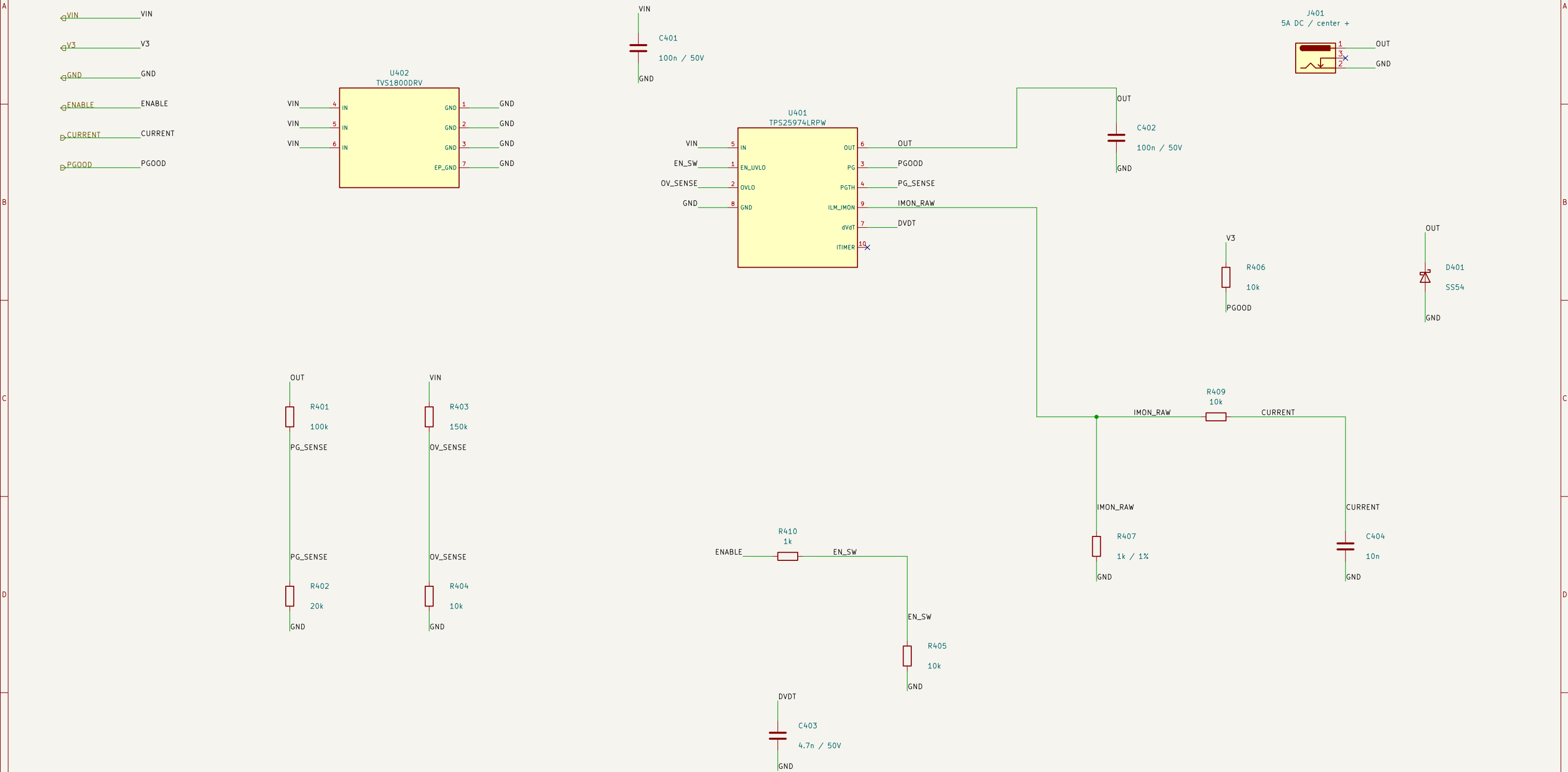


TPS25974L: 1k sets 5.75 A nominal breaker threshold for 5 A service. ITIMER open: latch-off.
ILM is BOTH current setting and monitoring: 105.5 uA/A x 1k = 0.1055 V/A; 0.528 V at 5 A.
10k/10n filter isolates ADC capacitance from ILM. Calibrate channels; do not attach a capacitor directly to ILM.
Independent local OVLO: 150k/10k -> 19.2 V nominal. PG threshold: 100k/20k -> 7.2 V nominal.
Positive line switched; ground permanently connected. No reverse blocking; do not backfeed outputs.
IN pad5 / OUT pad6 are power/thermal rails, NOT GND. Reserve copper and thermal-via space.
TVS1800 flat clamp replaces SMBJ18A: 24.7 V max at specified 35 A / 8–20 us / 125 C pulse; verify loop overshoot.

USB DOWNSTREAM | 500 mA service | independent power and overcurrent

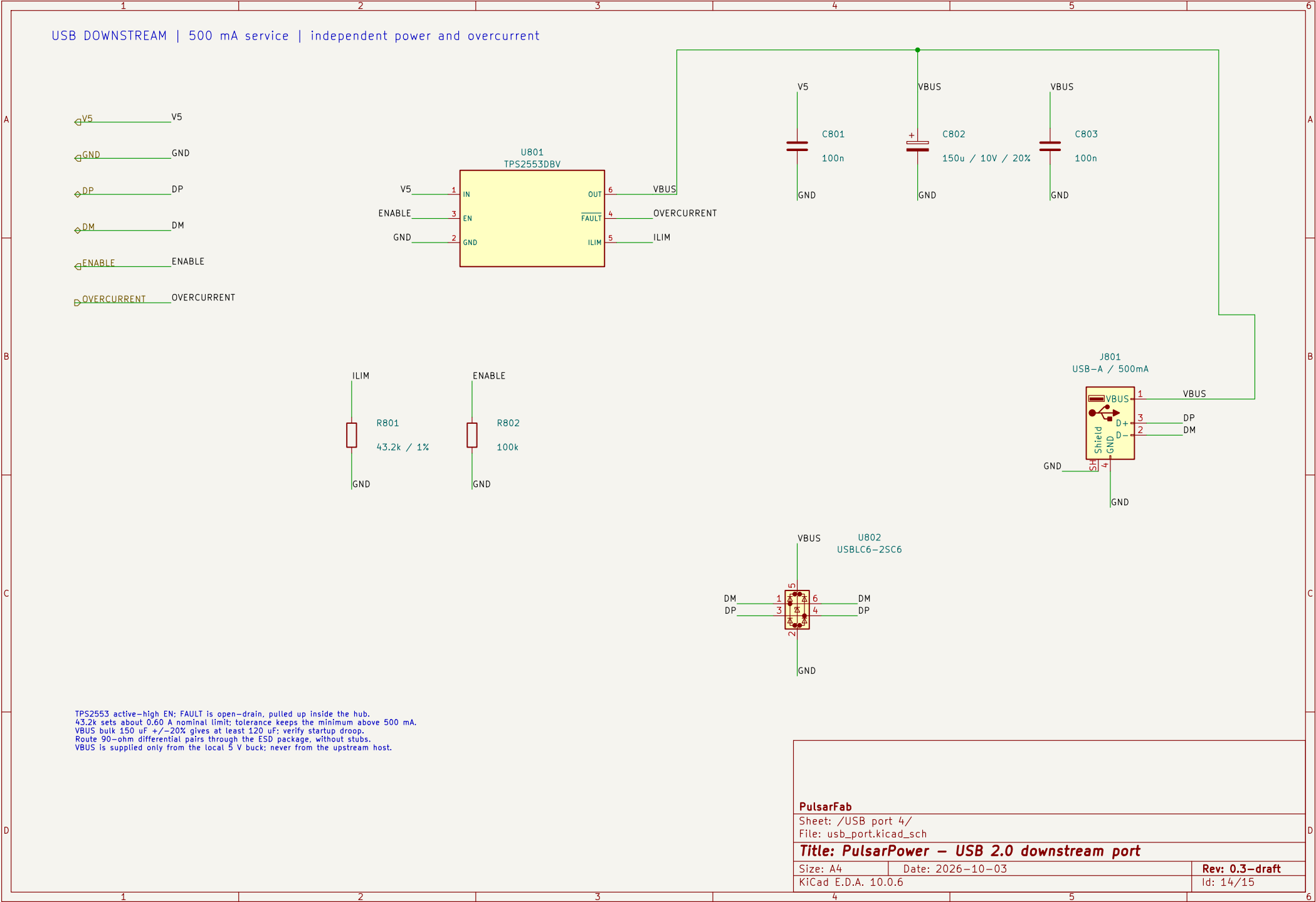


SWITCHED DC OUTLET | 12–18 V / 5 A service | hardware voltage inhibit / latched breaker

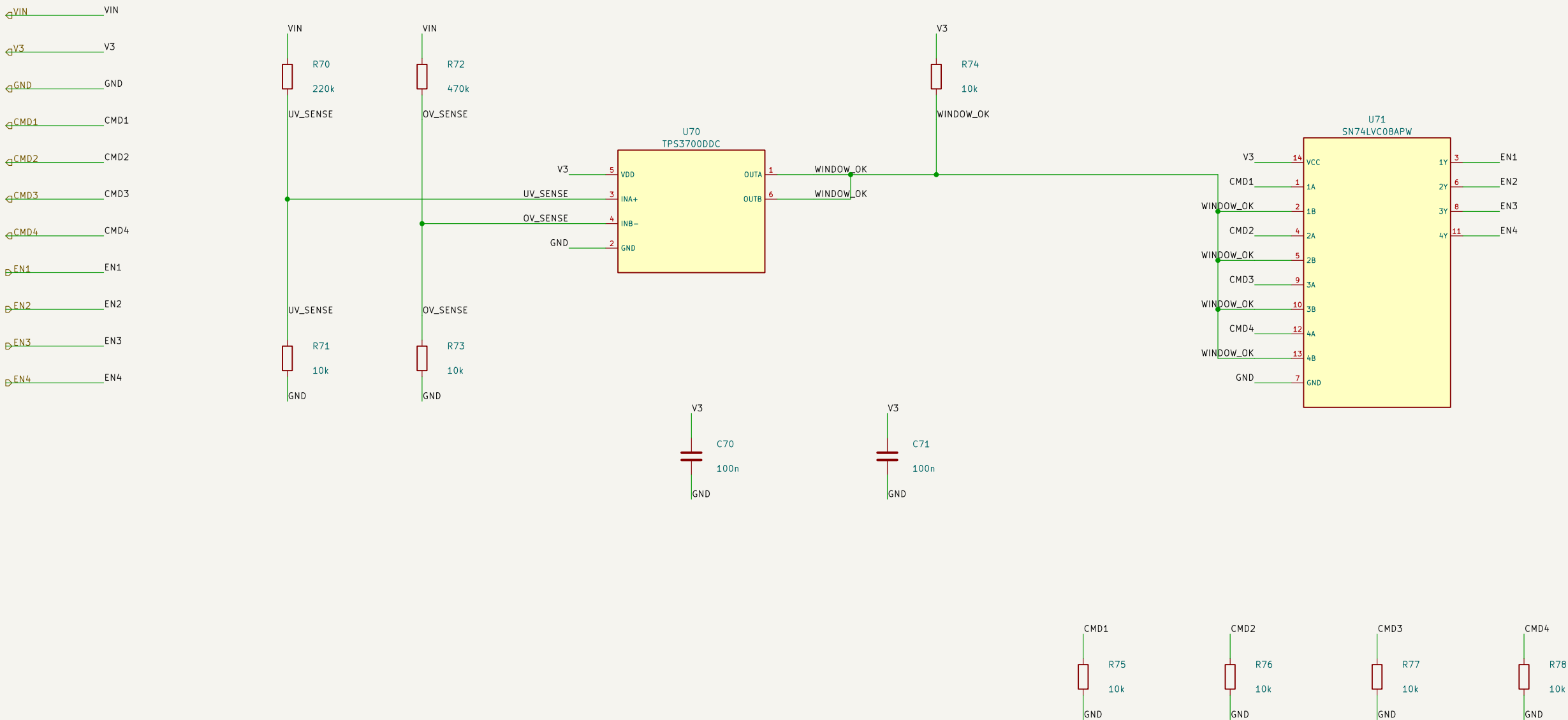


TPS25974L: 1k sets 5.75 A nominal breaker threshold for 5 A service. ITIMER open: latch-off.
ILM is BOTH current setting and monitoring: 105.5 uA/A x 1k = 0.1055 V/A; 0.528 V at 5 A.
10k/10n filter isolates ADC capacitance from ILM. Calibrate channels; do not attach a capacitor directly to ILM.
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TVS1800 flat clamp replaces SMBJ18A: 24.7 V max at specified 35 A / 8–20 us / 125 C pulse; verify loop overshoot.

USB DOWNSTREAM | 500 mA service | independent power and overcurrent



DC PROTECTION | one hardware voltage window | four independent default-off commands



TPS3700: 9.2 V UV rising / 19.2 V OV rising nominal. Open-drain outputs wired together.
74LVC08: ENi = CMDi AND WINDOW_OK; each command has a 10k pulldown.
Each outlet also has an EN pulldown: reset, missing logic supply and disconnected command request OFF.
A voltage recovery can re-enable an asserted command and reset a latched breaker; firmware must deassert on unexpected PG loss.
Shared inhibit is a common failure point, not a redundant safety channel. Local eFuse OVLO remains independent.
This circuit does not disconnect VIN from the regulator, heater stages or eFuse input pins.